The bridges of EDAM county

Integrating Bioconductor packages with the ELIXIR Research Software Ecosystem using EDAM

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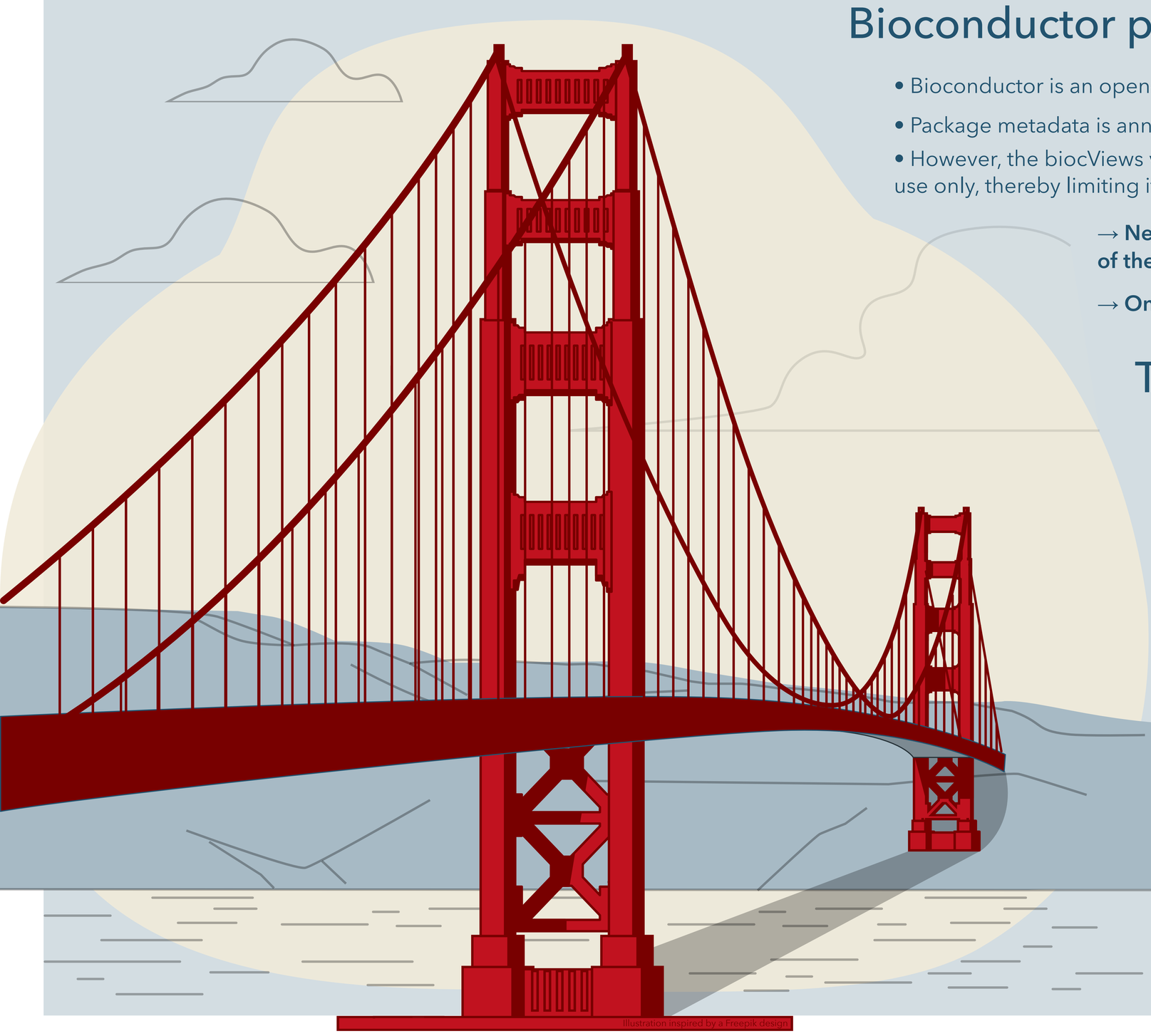


Illustration inspired by a Freepik design

Bioconductor packages for bioinformatics

- Bioconductor is an open-source project[a] with a collection of 2,000 software packages
- Package metadata is annotated with the biocViews[b] controlled vocabulary
- However, the biocViews vocabulary is not formally defined & is intended for internal use only, thereby limiting its interoperability with other resources

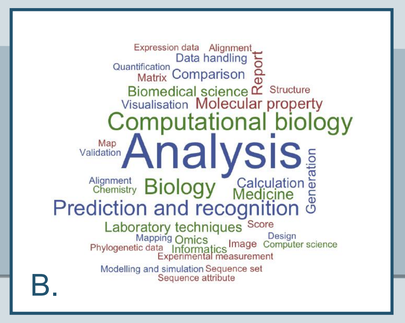
→ Need for standardised metadata to ensure the interoperability of the Bioconductor package collection

→ Ontologies can help achieve this standardisation

The EDAM ontology

- Definition of concepts for the annotation of data analysis and management resources in the field of life sciences[c]
- 4 main classes of concepts: topic, operation, data, format
- EDAM is currently a key component for ELIXIR resources and services interoperability

→ EDAM could be used as a bridge between the Bioconductor package collection & the ELIXIR ecosystem



A. Most used biocViews terms in Bioconductor package metadata

B. Top-level EDAM topics (green), operations (blue) and data types (red)

Building interoperable bridges

Vocabulary mapping

- Mapping of biocViews terms to EDAM topics using the text2term python library[e]
- Resulting matches manually curated and non-relevant matches revised (Figure C.)

→ 250 mapped and curated terms, 181 matches (72.4%) including 53 modified matches (21.2%)

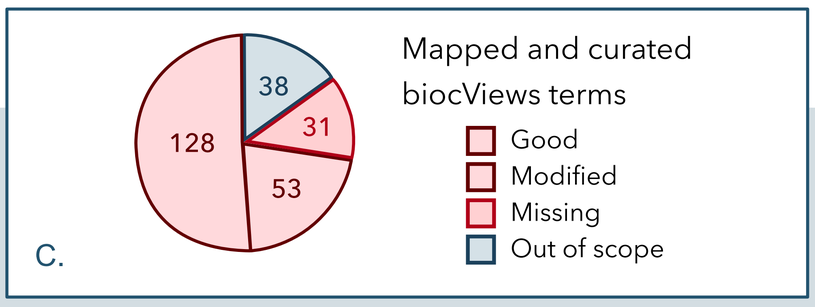
→ 29 new EDAM concepts suggested for 31 biocViews terms with no current match (12.4%)

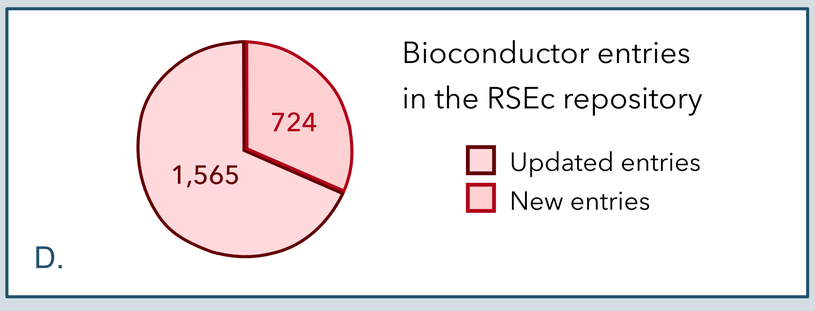
Metadata synchronisation

- Automated weekly synchronisation of Bioconductor metadata with both the bio.tools registry[f] and the ELIXIR Research Software Ecosystem (RSEc)[d] (Figure D.)

→ 1,565 updated entries and 724 new entries in RSEc (soon in bio.tools)

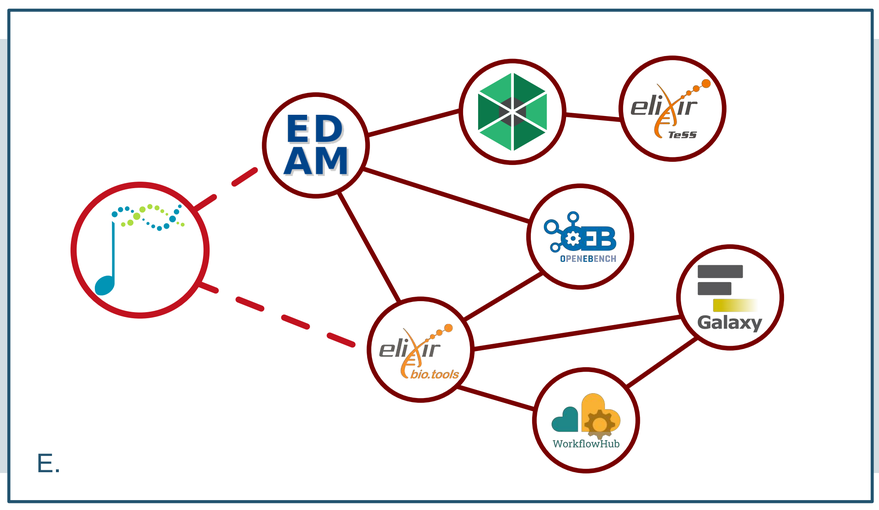
→ More detail and results in our recently published 2024 ELIXIR BioHackathon project preprint[i]





C. Mapping & curation of biocViews terms to EDAM concepts

D. Synchronisation of Bioconductor package metadata with the RSEc



Conclusion & perspectives

→ Together, our results establish bridges between the Bioconductor project and the RSEc, with further interoperability with many resources (Figure E.)

- Aim 1: Further develop LLM-based approaches to automate package annotation
- Aim 2: Extend metadata with EDAM operations, data and formats
- Aim 3: Initiate the vocabulary switch in concertation with Bioconductor developers

→ Project submitted for the 2025 ELIXIR BioHackathon with the Metabolomics community

E. Overview of some of the main interoperable resources and services within the ELIXIR ecosystem

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References

a. Bioconductor project. Huber et al. (2015), doi.org/10.1038/nmeth.3252
b. biocViews vocabulary. Carey et al. (2024), bioconductor.org/packages/biocViews.
c. EDAM ontology. Ison et al (2013), doi.org/10.1093/bioinformatics/btt113
d. ELIXIR Reasearch Software Ecosystem. github.com/research-software-ecosystem
e. Text2term library. rsgoncalves.com/text2term
f. bio.tools registry. Ison et al. (2019), doi.org/10.1186/s13059-019-1772-6
g. Lobentanzer et al. (2025), doi.org/10.1038/s41587-024-02534-3
h. Biochatter module. https://github.com/biocypher/biochatter
i. BioHack24 report. Rioualen et al. (2025), QRcode on the right